FUNCTORS SATISFYING THE COVERING CONDITIONS DEFINE POINTS OF THE TOPOS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The adjunction between the direct image and the stalk functor. Same chapter, Lemma lemma-adjoint-point-push-stalk. Supplies the adjointness restricted to sheaves.
- The universal property of sheafification. Same chapter, Proposition proposition-sheafification-adjoint. Supplies the adjunction used in the final computation.

Source reference. Original lemma 00Y7: sources/stacks-sites/pinned-original-source.tex, source lines 8223–8223. Navigation pointer invoked at proof line 8223; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p = u : \mathcal{C} \to Sets$ be a functor. Suppose that for every covering $\{U_i \to U\}$ of $\mathcal{C}$*

(1) *the map $\coprod u(U_i) \to u(U)$ is surjective, and*
(2) *the maps $u(U_i \times_U U_j) \to u(U_i) \times_{u(U)} u(U_j)$ are surjective.*

Then we have

(1) *the presheaf $u^p E$ is a sheaf for all sets E, denote it $u^s E$,*
(2) *the stalk functor $Sh(\mathcal{C}) \to Sets$ and the functor $u^s : Sets \to Sh(\mathcal{C})$ are adjoint, and*

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(3) *we have* $\mathcal{F}_p = \mathcal{F}_p^\#$ *for every presheaf of sets* $\mathcal{F}$.

Proof. The first assertion is immediate from the definition of a sheaf, assumptions (1) and (2), and the definition of u^pE. The second is a restatement of the adjointness of u^p and the stalk functor (Lemma 00Y7) restricted to sheaves. The third assertion follows as, for any set E, we have

$$\mathrm{Map}(\mathcal{F}_p, E) = \mathrm{Mor}_{PSh(\mathcal{C})}(\mathcal{F}, u^pE) = \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{F}^\#, u^sE) = \mathrm{Map}(\mathcal{F}_p^\#, E)$$

by the adjointness property of sheafification. □